Chapter 4 Practice Quiz: Answer Key

1. A
2. D
3. C
4. A
5. D
6. B
7. A
8. C
9. B
10. D
11. D (40 N, the peak of the graph)
12. B (the box is not sliding, so static friction equals the push)
13. (a) 25 N (kinetic friction, from the graph)
    (b) 55 N - 25 N = 30 N
    (c) a = F/m = 30 N / 10 kg = 3 m/s^2
14. C (speed triples, so $3^2 = 9$ times)
15. B
16. C
17. A
18. C
19. D
20. B
21. A
22. m = F/a = 18 N / 3 m/s^2 = 6 kg
23. 500 g = 0.5 kg. F = ma = 0.5 kg $\times$ 6 m/s^2 = 3 N
24. (a) a = change in v / t = 20 m/s / 10 s = 2 m/s^2
    (b) F = ma = 1000 kg $\times$ 2 m/s^2 = 2000 N
25. (a) W = mg = 40 kg $\times$ 10 N/kg = 400 N
    (b) W = 40 kg $\times$ 1.6 N/kg = 64 N
    (c) 40 kg (mass does not change)
26. (a) 25 N - 5 N = 20 N to the right
    (b) a = F/m = 20 N / 4 kg = 5 m/s^2 to the right
27. (a) W = mg = 2 kg $\times$ 10 N/kg = 20 N
    (b) net force = 20 N - 8 N = 12 N (downward)
    (c) a = F/m = 12 N / 2 kg = 6 m/s^2 (downward)
28. Speed triples, so air resistance is $3^2 = 9$ times as big: 9 $\times$ 20 N = 180 N
29. (a) W = mg = 3 kg $\times$ 3.7 N/kg = 11.1 N
    (b) a = F/m = 11.1 N / 3 kg = 3.7 m/s^2 (the same as g)

30. (a) 1.4 + 0.6 = 2.0 kg
    (b) Pulling force = weight of the hanger = 0.6 kg × 10 N/kg = 6 N
    (c) a = 6 N / 2.0 kg = 3 m/s^2
31. (a) Pulling force = weight of the hanger = 0.4 kg × 10 N/kg = 4 N
    (b) total mass = F/a = 4 N / 2 m/s^2 = 2.0 kg
    (c) cart = 2.0 kg - 0.4 kg = 1.6 kg